Refraction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect a change in wave speed to wavelength and direction.

SUCCESS CRITERIA - I can...

1. Measure ray angles from the normal.
2. Explain frequency continuity at a stationary boundary.
3. Predict bending toward or away from the normal.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “refraction” and “normal line” to explain your answer. Sentence starters are optional.

refraction: change in wave direction when speed changes at a boundary

normal line: a reference line perpendicular to a surface

medium: the material through which a mechanical wave travels

01 Warm-up: bring back an earlier idea

A wave has period **3 s**. **Find the cycles completed** in **15 s** and its frequency.

Word help: frequency = number of cycles each second

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

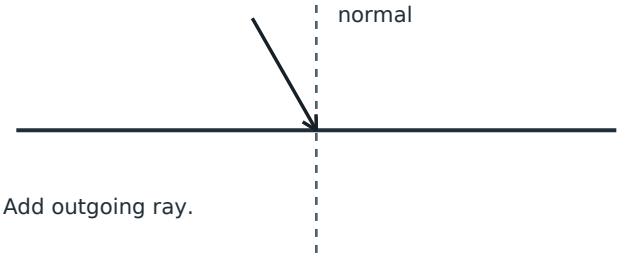
02 Worked example: explain the prediction

Reference: $v_1 = f\lambda_1$; $v_2 = f\lambda_2$; $\lambda_2/\lambda_1 = v_2/v_1$.
Stationary boundary: frequency stays constant. A slower medium bends an oblique ray toward the normal.

Worked reasoning: Light enters a slower medium at an angle to the normal and bends toward the normal. The normal is perpendicular to the boundary.

Your turn: Draw and **label the boundary**, normal, incident ray and refracted ray.

Sentence starter: In my work, I used … to show … . I labeled it … .

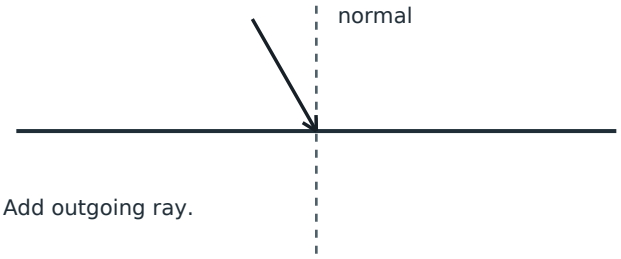


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Mark the normal at the boundary. A ray enters a slower medium at an angle. **Sketch the refracted direction**.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

04 Apply the physics

A wave crosses into a slower medium. Its speed changes from **12** to **6 m/s** while frequency stays **3 Hz**.
1. **Find the wavelength** on each side.
2. **Draw wavefronts** with the correct relative spacing.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student measures an incidence angle from the surface instead of the normal. If surface angle is **20°**, **what angle** belongs in Snell’s law?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Samples at **14°C** and **56°C** are mixed with no energy input or phase change. A student reports **68°C** at equilibrium. Is that possible? Explain.
Label temperatures, samples and relevant energy transfers.

Sentence starter: I think … . I can point to …, which shows …, so … .

Sample A

Sample B

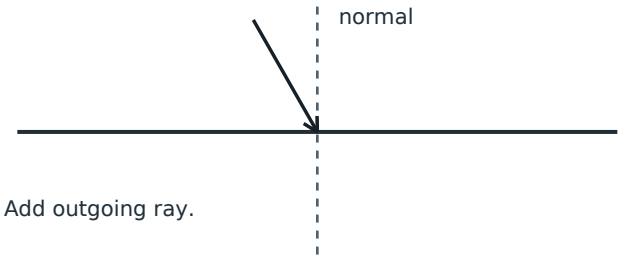
Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

A ray enters a faster medium at an angle. Does it bend toward or away from normal?

Sentence starter: I notice … . This supports … because … .



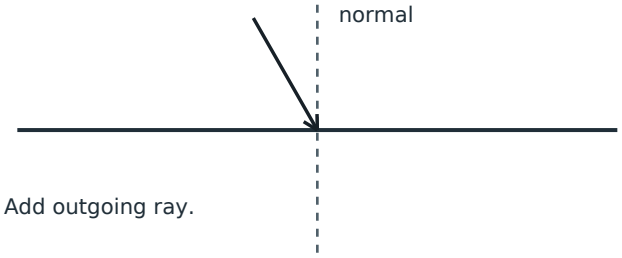
Add outgoing ray.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. **Draw two rays entering a** slower medium: one obliquely and one along the normal.
2. **Explain why one changes direction** and the other does not.

Sentence starter: I think … . I can point to …, which shows …, so … .



Add outgoing ray.

Marzano check - circle: 1 2 3